COMP S492F/3920SEF Machine Learning

Supervised machine learning

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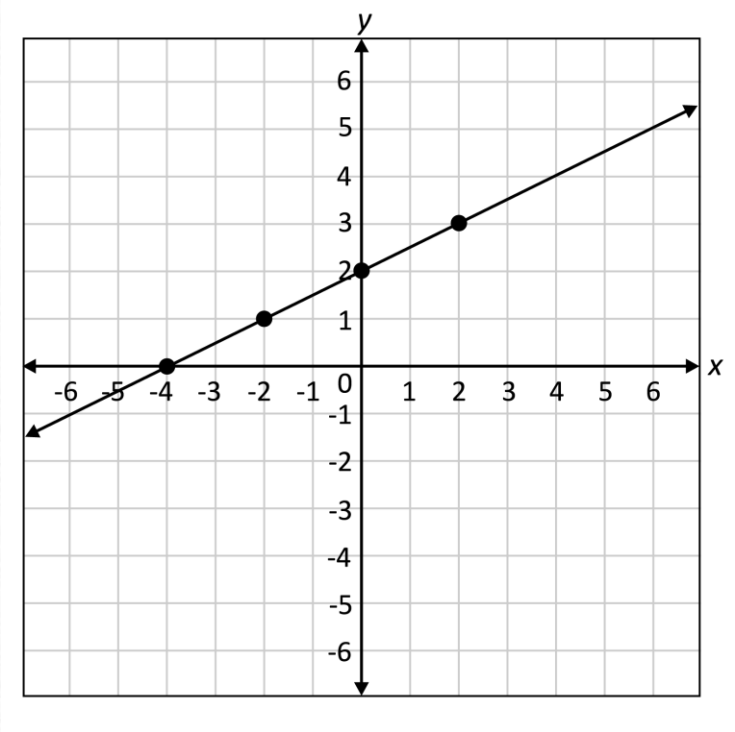
Linear and Logistic Regression

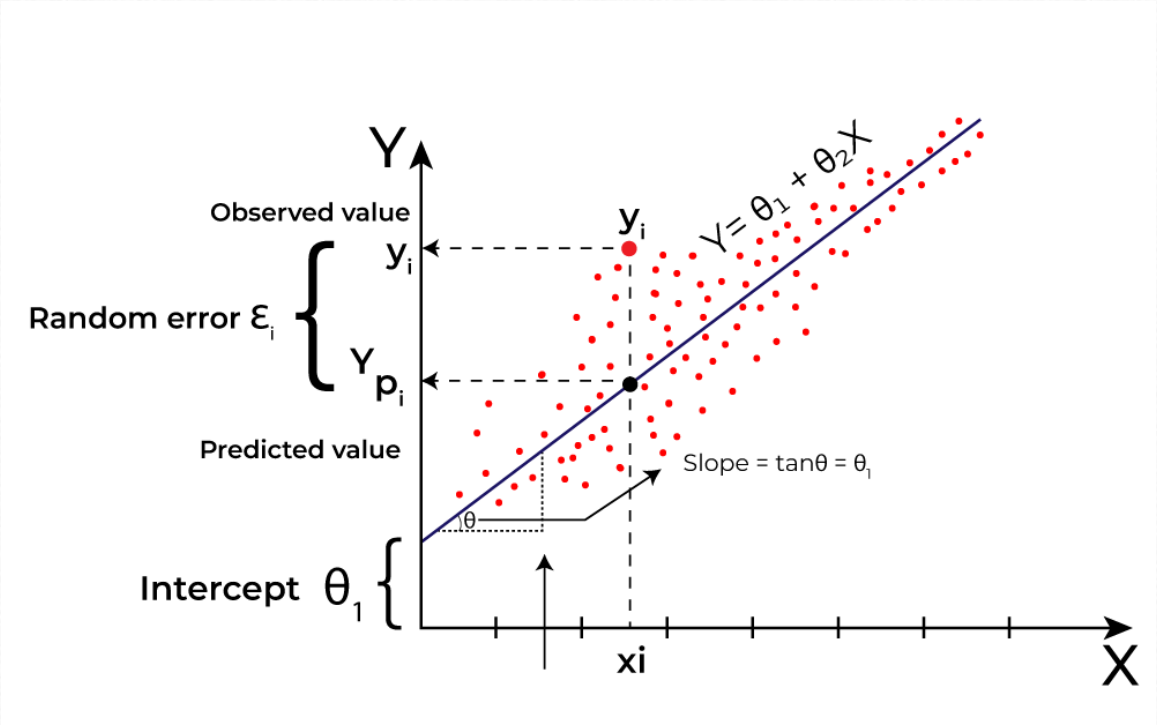
Guess what is the last number of each row?

1	2	3	4	5	6	7
1	3	5	7	9	11	?
2	4	6	8	10	12	?
1	5	9	16	25	36	?
2	4	8	16	32	64	?

Are you able to write the formular for each row

Linear and Logistic Regression





Linear and Logistic Regression

Linear Regression

- Predict an output which is a real number (i.e. continuous), e.g., prediction of weight according to height or more complex things like prediction of stock or property prices.
- In linear Algebra you teacher might tell you that linear regression has two assumptions:
  - 1) The input data and output data have a linear relationship;
  - 2) The random noise has a normal (Gaussian) distribution.

Forget about these in machine learning

In machine learning, linear means linear in parameters

$y = \beta_0 + \beta_1 x$ → linear regression you have learnt
$y = \beta_0 + \beta_1 x + \beta_2 x^2$ → Polynomial but linear in β's
$y = \beta_0 + \beta_1 \log(x) + \beta_2 \sqrt{x}$ → Nonlinear transforms of x, but linear in β's
$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$ → Interaction term, but linear in β's

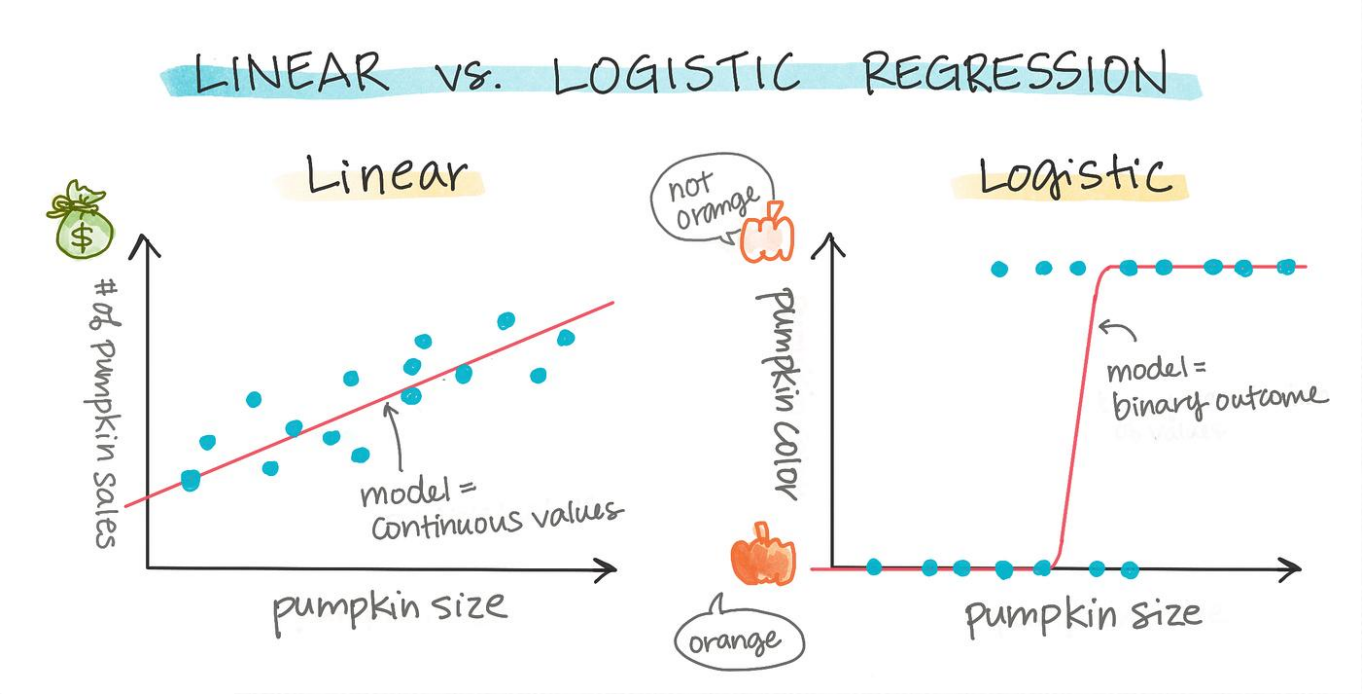
The relationship between x and y are not linear but they belongs to linear regression in machine learning

Linear and Logistic Regression

Logistic Regression

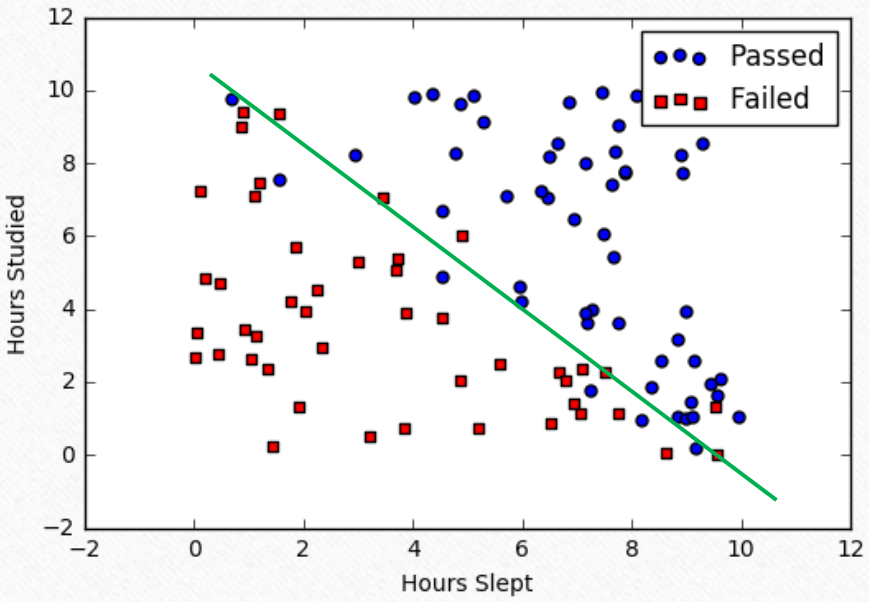
- Predict an output which is a categorical value (i.e., discrete), e.g. prediction of diabetes in patients.
- Estimate the probability of each category
- Assume a particular functional form: sigmoid.

Attention! Logistic Regression is a classification method



How Logistic Regression Works

In logistic regression, a linear equation with optimized coefficients is to be found to give a decision boundary (in green) which can be used to classify the featured data into 2 classes as most accurate as possible (e.g., to predict a student to get a Pass or failure with 2 features: hours slept and hours studied).



Steps taken (for binary classification):

1. Compute weighted outputs using a linear model equation (similar to linear regression).
2. Transform its output using the **Sigmoid function** to return a probability value which can then be used for mapping to the discrete classes.
3. Calculate the **Cross Entropy** of the predicted output and use it as the cost function.
4. Update the coefficients iteratively by using **gradient descent** as an optimization algorithm to minimize the cost function.
5. Stop until convergence or an acceptable error limit is reached.

OMG a lot of terminologies, you will learn about them in the following classes

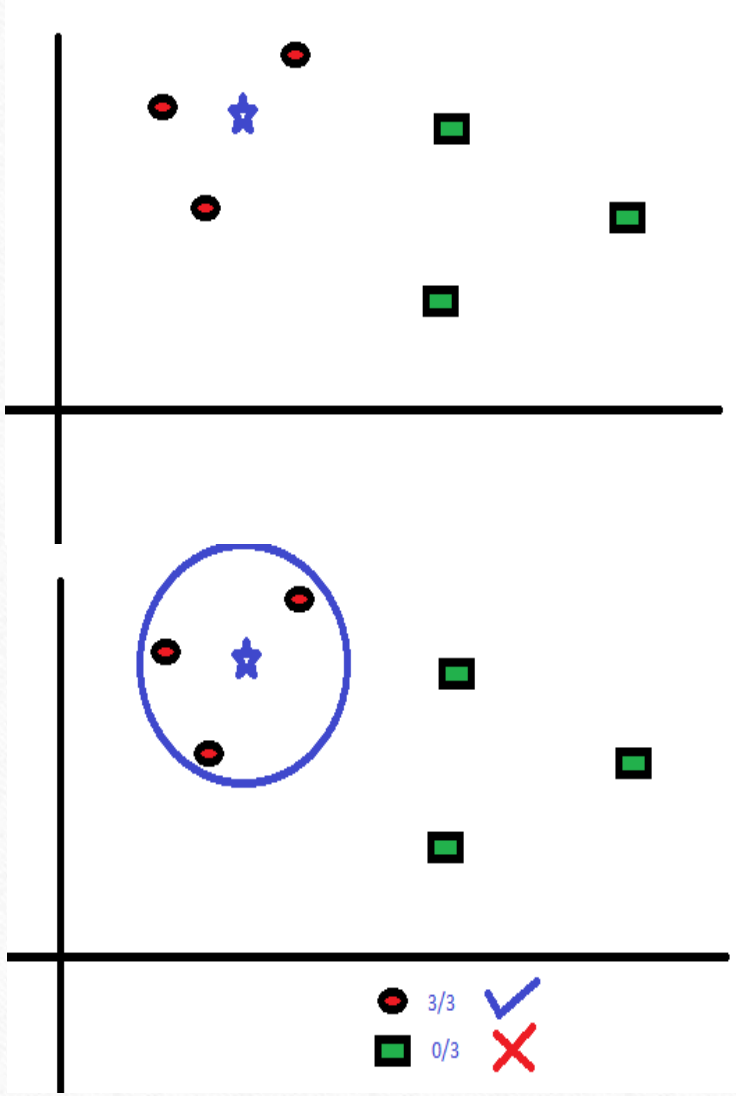
K-Nearest Neighbors

K-Nearest Neighbors (KNN):

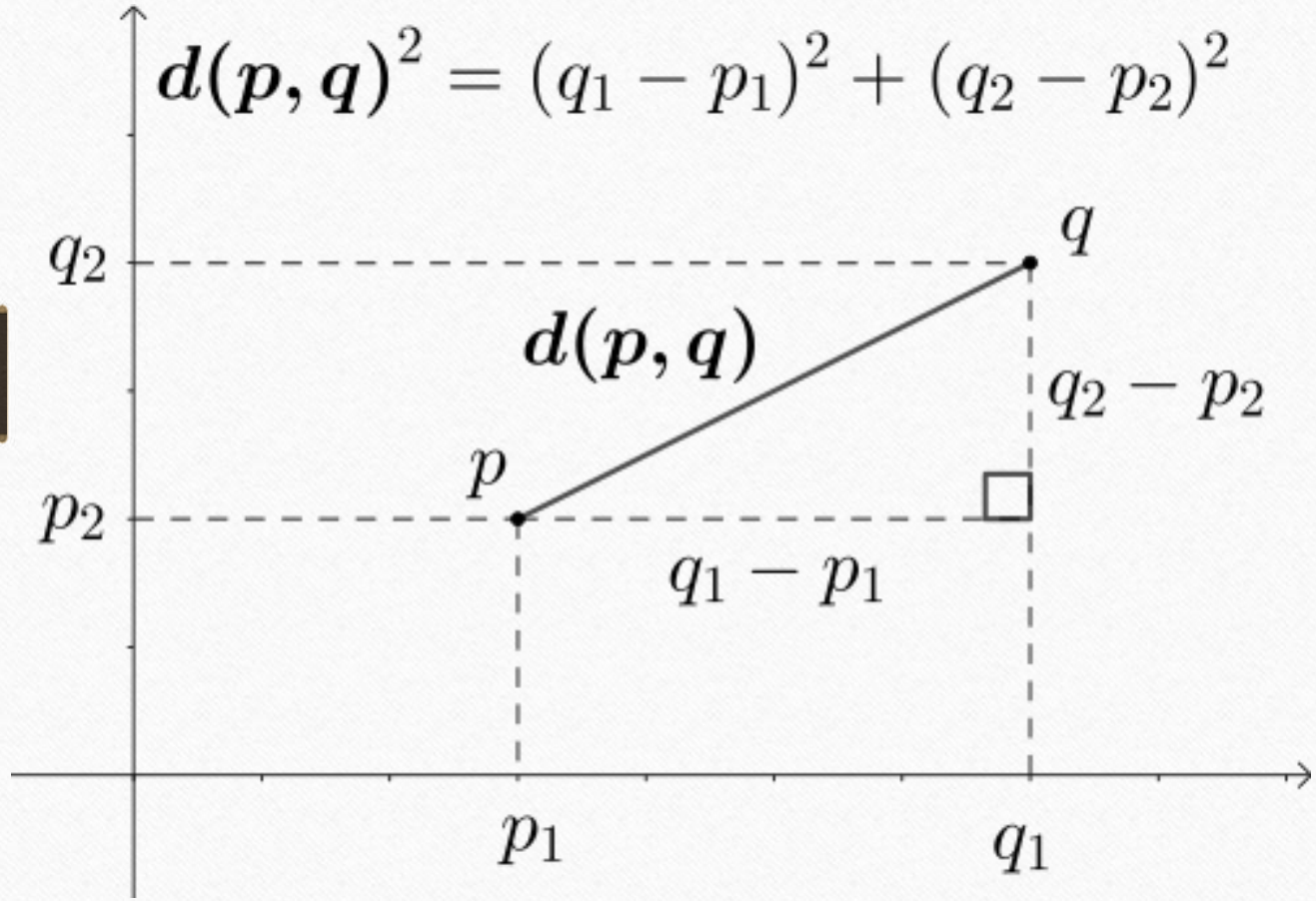
- It is one of the simplest ML algorithms and is an example of instance-based learning (memory-based learning), where new data are classified based on stored and labeled instances.
- There is no training step. The model is just the whole dataset.
- KNN is usually used in classification but can also be used in regression.

How K-Nearest Neighbors Works (Example)

- The figure shows a spread of red circles (RC) and green squares (GS). You intend to find out the unknown class of the blue star (BS), which can be either RC or GS.
- The “K” in KNN algorithm is the number of the nearest neighbors we wish to take vote from. Let’s say K = 3.
- Hence, we will now make a circle with BS as center just as big as to enclose only three data points on the plane. The metric often used for distance measurement is the Euclidean distance, i.e., $dist(a,b) = \sqrt{(a_1 - b_1)^2 + \cdots + (a_n - b_n)^2}$.
- The three closest points to BS are all RC. By voting with a result 3/3 votes (i.e. 100%) for RC, we can say that the BS should belong to the class RC.



How to calculate Euclidean Distance?
Based on Pythagorean theorem



Euclidean Distance

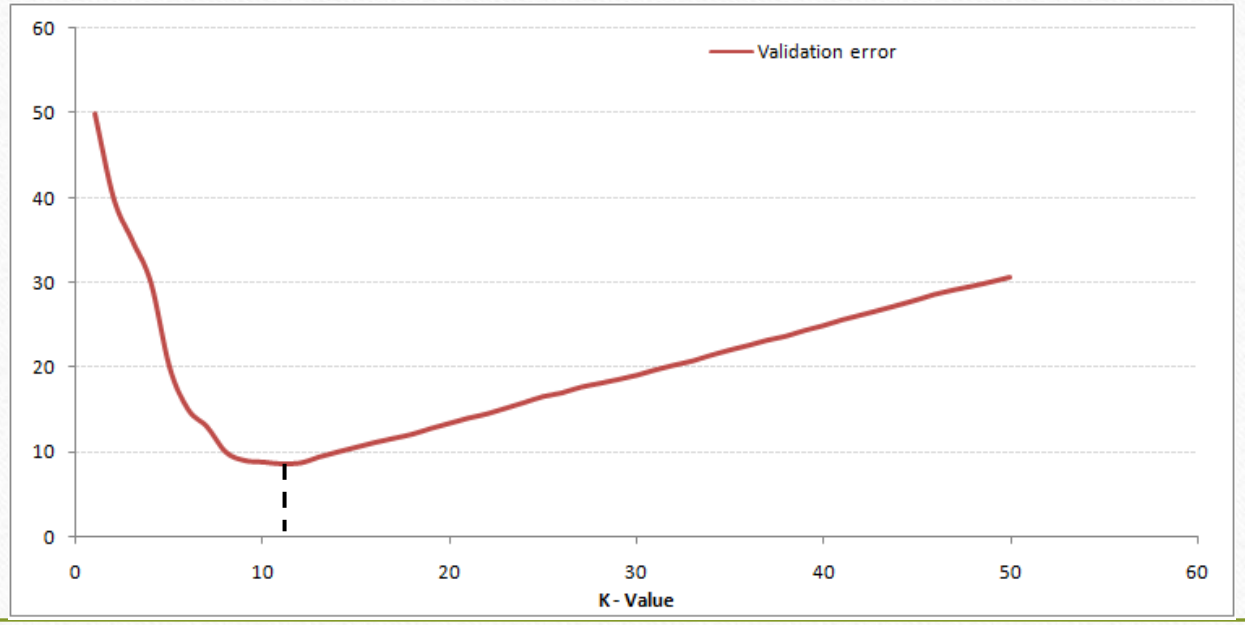
$$d = \sqrt{(q_1 - p_1)^2 + (q_1 - p_1)^2}$$

Choice of K

- To get the optimal value of K, one way is to split the training and validation sets from the initial dataset, e.g., in a ratio of 3 : 1. The aim is to use the training set to predict for the validation set.
- Use the classification inaccuracy as a measure of *validation error*:

  Validation error % = (No. of examples wrongly predicted/Total no. of examples) x 100%

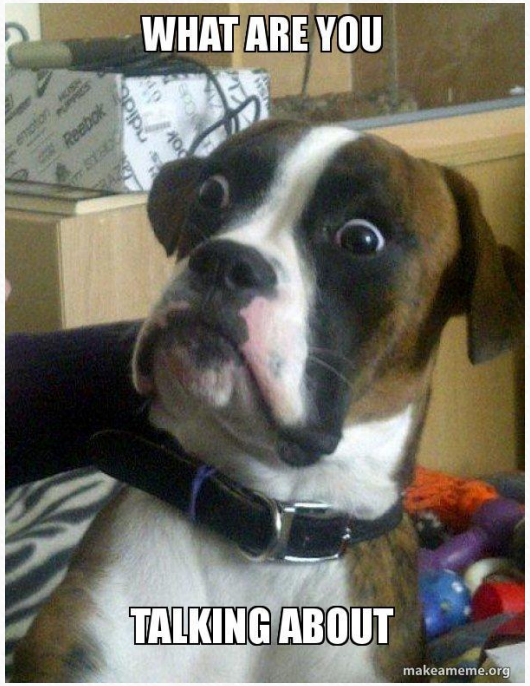
- Plot the validation error curve and locate its minimum to get the optimal value of K. This value of K should be used for all predictions.



Support Vector Machine

Support vector machine (SVM):

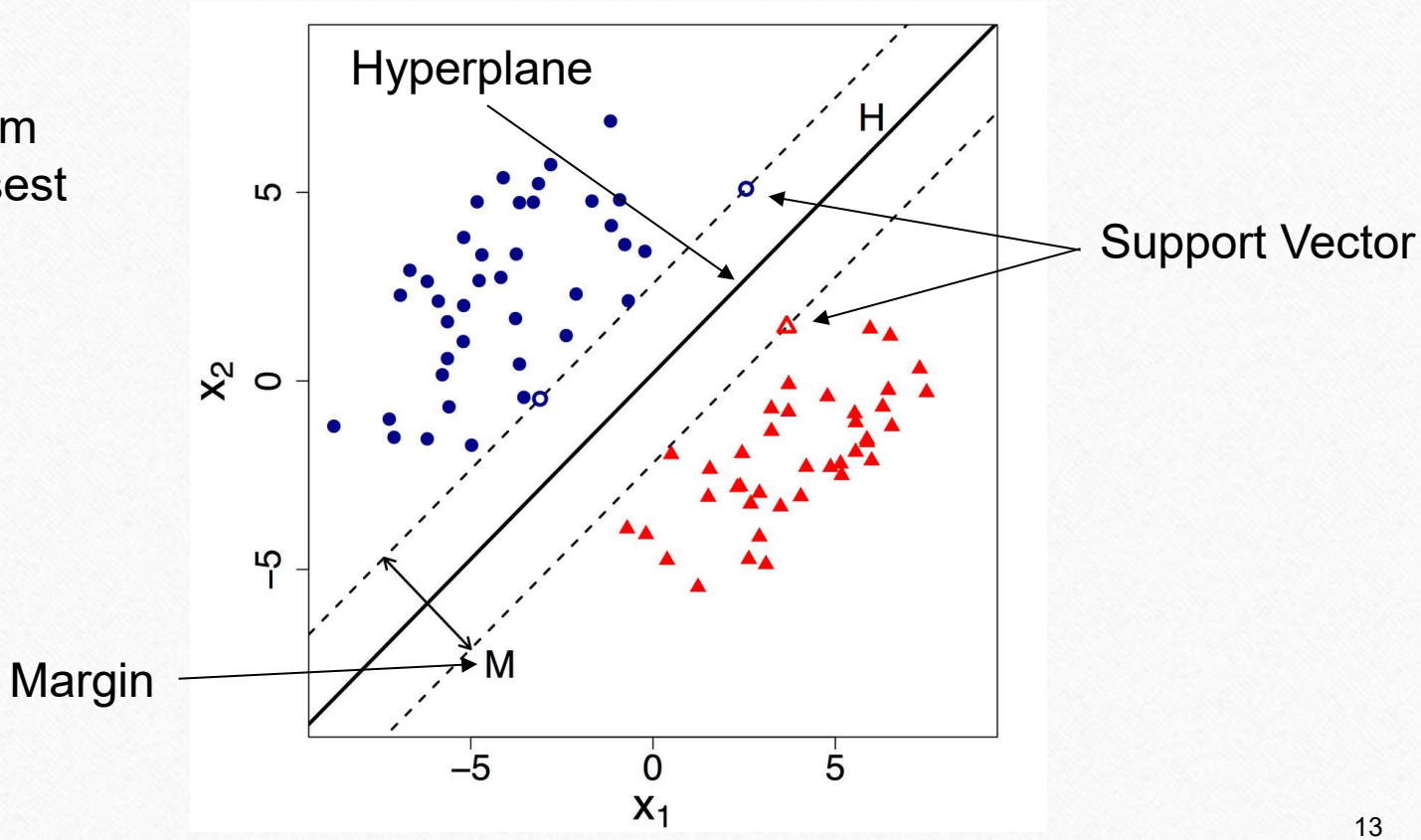
- Supervised machine learning algorithm used to find an optimal **hyperplane** (boundary) that best separates data points of different categories.
- SVM aims to find the decision boundary with the **maximum margin** between classes, which improves generalization and reduces overfitting.
- Only the training samples closest to the hyperplane (support vectors) influence its position; other points are ignored
- SVM can handle **non-linearly separable data** by mapping inputs into higher-dimensional spaces using **kernel functions**



Support Vector Machine

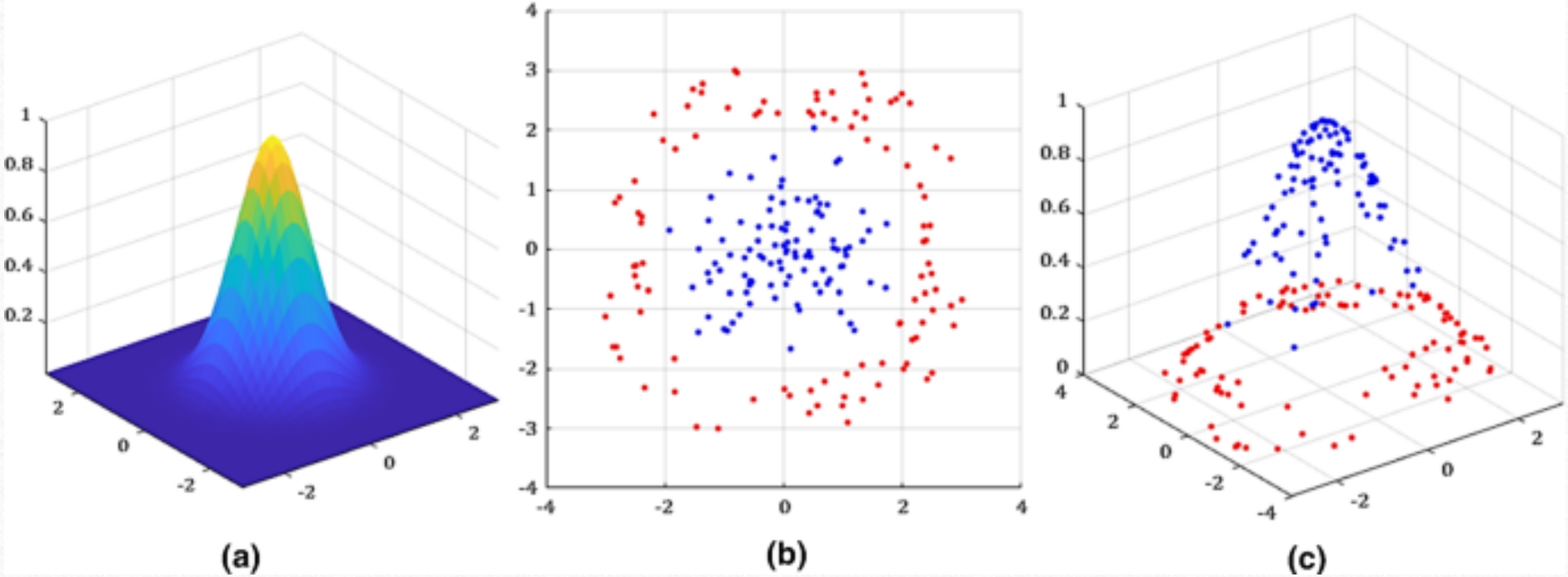
Find a boundary with maximum margin, only the samples closest to the hyperplane matters

What if the data is not linearly separable?



Support Vector Machine

Kernel Function | Original data | After kernel mapping



https://www.youtube.com/watch?v=_YPScrckx28

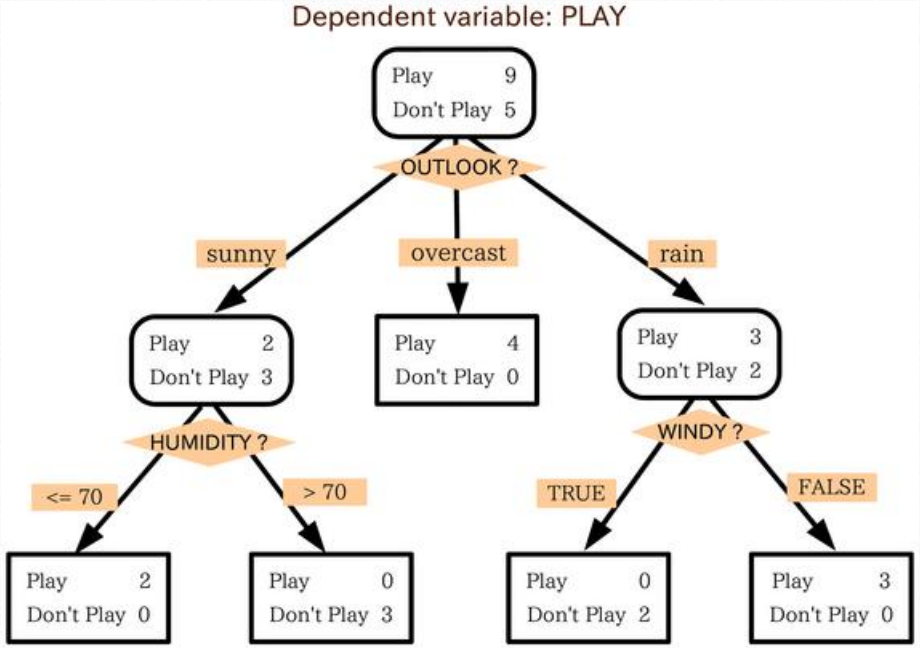
Random Forest

Random forest (RF):

- an **ensemble learning method** that constructs multiple **decision trees** during training and combines their predictions for classification
- It operates by building many trees on **random subsets** of the **data** and **features**, then aggregating their results with voting
- Each tree is trained on a **random sample** (sampling with replacement) of the training data, creating diversity.
- At each split, the algorithm considers only a **random subset of features**

Random Forest

Firstly what is a decision tree
e.g., you have difficulty in deciding playing tennis or not

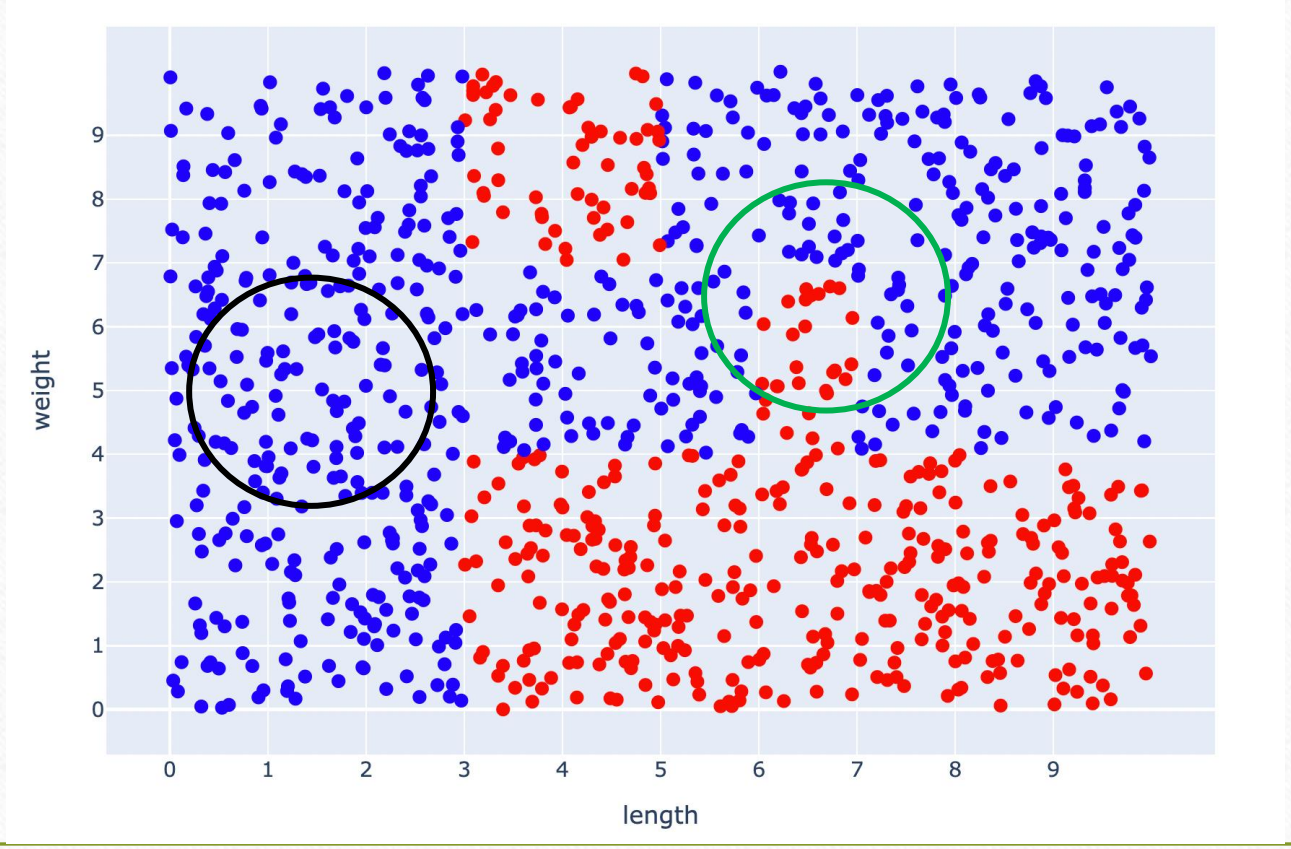


- **Internal nodes** represent features
- **Branches** represent outcomes of a split
- **Leaf nodes** represent final predictions

How Decision Tree decide the order and threshold of each node?

By information gain / decrease in entropy!

Or a simply speaking, it tries to minimize the uncertainty.



Which circle region do you think has higher entropy?

So, if you are a decision tree how will you do the first split?

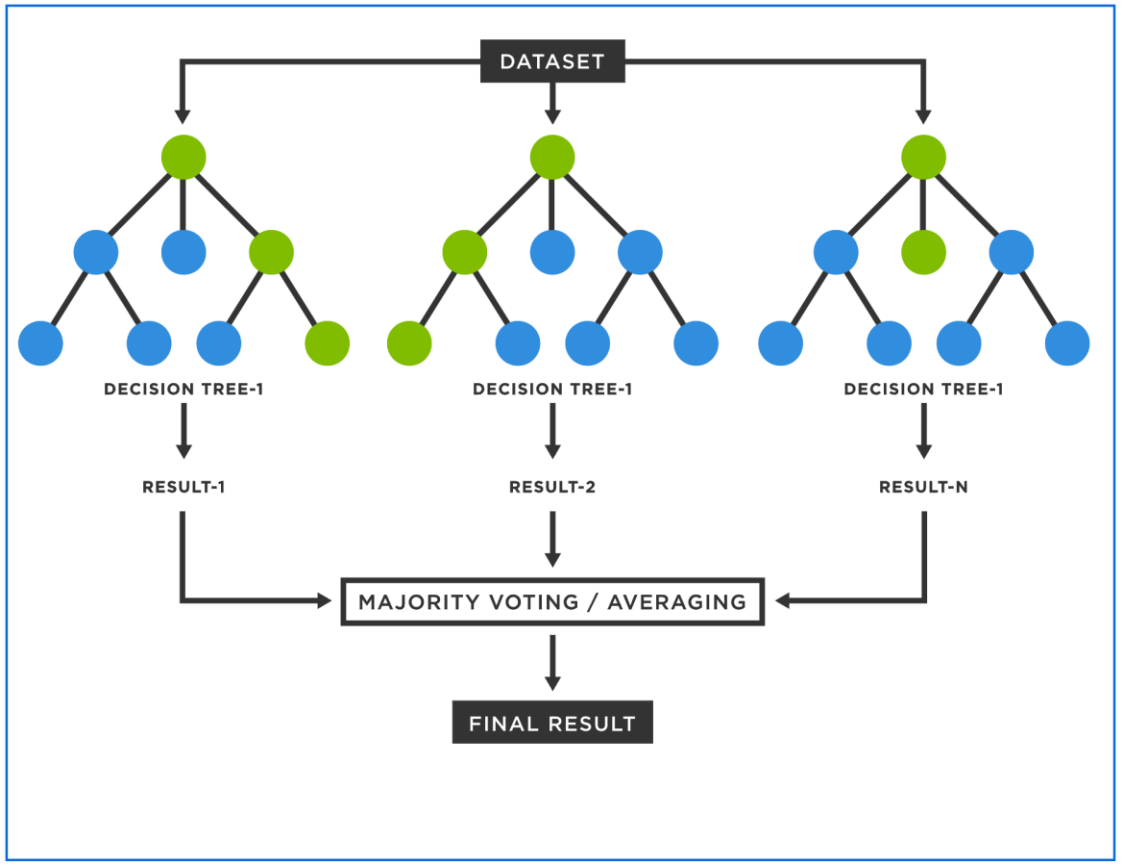
Random Forest

Then what is a Random Forest?

We know forest means a lot of trees!

So, the idea is that we build a lot of decision trees and each of them get a result, after which we get the final decision by majority voting or averaging among the trees

https://www.youtube.com/watch?v=v6VJ2RO66Ag



Pros and Cons of each Method

Algorithm	Key Strength	Key Weakness
Logistic Regression	• Interpretable - Coefficients show feature importance & direction • Fast training & prediction • Good starting point for classification problems	• Linear decision boundary only - Cannot capture complex non-linear patterns • for multi-class - Needs extensions
KNN	• No training phase • Incremental learning - Adding new data is trivial	• Slow prediction • Choice of k critical: Small k → overfitting, large k → underfitting
SVM	• Memory efficient • Versatile with kernels - Can model non-linear relationships via kernel	• Computationally expensive • Poor scaling • Hard to interpret (especially with kernels)
Random Forest	• High accuracy • Provides ranking of feature importance • Parallelizable - Trees can be built independently	• Computationally expensive • Memory intensive • robustness to outliers and noise